Chapter 1 Additional Text for Ring and Modules

1.1 Local Properties

1.1.1 Local Rings

1.1.2 Localization

1.2 Spectrum of Ring

Definition 1.2.0.1 Let R be CR. The spectrum is the topology given by:

$$Spec(R) := \{p \in R | p \text{ is prime}\}$$

1.2.1 Zariski Topology

Definition 1.2.1.1

1.2.2 Localization

1.2.3 Open and Closed Subsets of Spectra

Closed Points in $Spec(R)$

♠ **Proposition 1.2.3.1** Let $x = p \in Spec(R)$ be a point. Then

$$\{x\} \text{ is closed} \Leftrightarrow \text{ p is maximal ideal.}$$

∘ **Proof** $\{x\}$ is closed $\Leftrightarrow$ $\{x\} = V(x) = \{p \in Spec(R) | x \subseteq p\}$ $\Leftrightarrow$ x is a maximal ideal. □

1.2.4 Connected Component

1.2.5 Glueing Property

◇ **Lemma 1.2.5.1** ...

∘ **Proof** ... □

1.2.6 Glueing Functions

Goal: To show

$$Spec(R) = \cup_{i=1}^{n} D(f_i)$$

And given $h_i = h_j \in R_{f_i f_j}$, there exists $h \in R$ where $h = h_i \in R_{f_i}$.

- $D(f) \mapsto R_f$ defines a sheaf;
- If we think of the elements of R_f as the algebraic/regular functions on $D(f)$, then these glue as would continuous/differentiable functions on a topological/differentiable manifold.

◇ **Lemma 1.2.6.1** (Exactness of Sheaf of Modules)

∘ **Proof** ... □

◇ **Lemma 1.2.6.2** Let R be a ring, $f_1, \dots, f_n \in R$ generarte the unit ideal in R, i.e.

$$R = (1) = (f_1, \dots, f_n)$$

Then, the following sequence is exact:

$$0 \to M \to \oplus M_{f_i} \to \oplus_{i,j} M_{f_i f_j} \to 0$$

Given by:

$$\alpha : M \to \oplus M_{f_i} \qquad x \mapsto (\frac{x}{1}, \dots, \frac{x}{1})$$
$$\beta : \oplus M_{f_i} \to \oplus M_{f_{ij}} \qquad (\frac{x_1}{f_1^{r_1}}, \dots, \frac{x_n}{f_n^{r_n}}) \mapsto (\frac{x_i}{f_i^{r_i}} - \frac{x_j}{f_j^{r_j}})_{i,j}$$

∘ **Proof**

- Injectivity: We have

$$(\frac{x}{1}, \dots, \frac{x}{1}) = 0 \Leftrightarrow f_i^{e_i} \cdot x = 0; e_i \geq 1$$

And notice that,

$$1 = \sum \alpha_i f_i \Rightarrow x = (\sum \alpha_i f_i)^{\max\{e_i\}} \cdot x = 0$$

- Surjectivity: Consider arbitrary

$$s = \frac{z}{(f_i f_j)^r} \Rightarrow \frac{x_i f_j^{r_i} - x_j f_i^{r_i}}{f_i^{r_i} f_j^{r_j}} = \frac{(x_i^{r_i} f_j^{r_i} - x_j f_i^{r_i}) f_i^{r-r_i} f_j^{r-r_j}}{(f_i f_j)^r} \qquad r = \max\{r_i, r_j\}$$

There exists construction:

$$z = x_i f_j^r f_i^{r-r_i} - x_j f_i^r f_j^{r-r_j}$$

And with the discussion of injectivity, we have

$$(f_i f_j)^s z = 0$$

Thus, we could construct as following:

$$x_i' := f_i^{r-r_i} x_i \qquad x_j' = f_j^{r-r_j} x_j$$

Then,

$$f_j^r$$

Finally, $(f_i^{r_i}) = R \Rightarrow \sum y_i f_i^r = 1 \Rightarrow x = \sum y_j x_j'$

□

★ **Corollary 1.2.6.3** Let R be a ring. Let $f_1, \dots, f_n \in R$. Let M be an R-module. Then, the following statements are equivalent:

- $M \to \oplus_i M, m \mapsto (f_1 m, \dots, f_n m)$ is injective;
- $M \to \oplus M_{f_i}$ is injective.

∘ **Proof** Notice that

$$M \to \oplus M_{f_i} \qquad inj. \Leftrightarrow f_i^{e_i} m = 0, i = 1, \dots, n$$

According to proof of part i above,⇔ Prove. □

Glueing Spectrum

◇ **Lemma 1.2.6.4**

- Suppose that
$$Spec(R) = U \sqcup V$$
where $U = Spec(R_1), V = Spec(R_2)$, then
$$R \cong R_1 \times R_2$$
- Moreover, if R_1 and R_2 are localizations as well as quotient of the ring R.

∘ **Proof**

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□

Pasting of Datas

♠ **Proposition 1.2.6.5** Let R be a ring, $f_1, \dots, f_n \in R$. Suppose we are given following data:

- For each i, and R_{f_i}-module M_i;
- For each pair i, j, an $R_{f_i f_j}$-module isomorphism
$$\psi_{ij} : (M_i)_{f_j} \to (M_j)_{f_i}$$

∘ **Proof** ... □

1.2.7 Irrducible Component

1.2.8 Image

Images of Ring Maps of Finite Representation

◇ **Lemma 1.2.8.1** Let $U \subseteq Spec(R)$ be open. The followings are equivalent:

- U is **retrocompact** in R;
- U is **quasi-comapct**
- U is finite union of standard opens;
- There exist a finitely generated ideal $I \subseteq R$ such that $X/V(I) = U$;

∘ **Proof**

- (3)⇔ (4) Omitted;
- (3)⇒ (4):
- (4)⇒ (1):
- (1)⇒ (3)

□

More on Images

1.3 Jacobson Radical of a Ring

1.3.1 Nakayama's Lemma

1.4 Additional Propositions for Prime Ideals

1.5 Nilpotent

Definition 1.5.0.1 Let R be a ring, $I \subseteq R$ be an ideal:

- I is **locally nilpotent** if for every $x \in I$, there exists n such that $x^n = 0$;
- I is **nilpotent** if there exists n such that $I^n = 0$;

◇ **Lemma 1.5.0.2** Let $R \to R'$ be a ring map, and $I \subseteq R$ be a locally nilpotent ideal, then IR' is a locally nilpotent ideal of R'.

∘ **Proof** □

◇ **Lemma 1.5.0.3** Let R be a ring and $I \subseteq R$ be a locally nilpotent ideal, an element x of R is a unit iff image of x in R/I is a unit.

∘ **Proof** □

◇ **Lemma 1.5.0.4**

∘ **Proof** □

1.6 Jacobson Ring

Definition 1.6.0.1

1.7 title

Chapter 2 Chain Conditions

Definition 2.0.0.1

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2.1 Noetherian Rings, Modules

2.1.1 Noetherian Ring

Definition 2.1.1.1

2.1.2 Noetherian Modules

Definition 2.1.2.1 Let R be a commutative ring, and M be $R{-}Mod$. Then, M is **Noetherian** if for any descending sequence of submodules

$$M_0 \subset M_1 \subset \dots \subset \dots$$

There exists some $N \geq 0$ such that $M_N = M_{N+1} = \dots$. i.e. The submodules satisfy ACC.

♠ **Proposition 2.1.2.2** The following conditions are equivalent:

- M is Noetherian;
- For every subset $S \subseteq M$, there exists elements $m_1, \dots, m_r \in S$,

$$(S)_R = (m_1, \dots, m_r)_R$$

- Every submodule of M is finitely generated.

◦ **Proof**

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□

♠ **Proposition 2.1.2.3** The following conditions are equivalent:

- M is Noetherian;
- M is finitely generated.

◦ **Proof**

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□

Polynomial Rings over Noetherian Rings

Given here:

2.2 Length

Definition 2.2.0.1 Let R be a ring. For any R-module M we define the **length** of M over R by the formula

$$length_R(M) := \sup\{\alpha \in Ord | M_0 = 0, M_\alpha = M, M_\gamma \subset M_{\gamma-1}\}$$

It is the superemum of legths of chain of submodules. There exists a partial order relation for chains by extending elements. Then the length of M is the length of maximal chain.

2.2.1 Properties

◇ **Lemma 2.2.1.1** $length_R(M) < \infty \Leftrightarrow M$ is a finite R-module.

- **Proof**
- $\Rightarrow$ We have

  $$M_i = (m_1, \dots, m_i)$$

  Then, the filtration is finite. Only finite generators.
- $\Leftarrow$ If $M = (m_1, \dots, m_n)$, then the maximal filration must be finite.

□

Lengths of Modules of short Exact Sequence

◇ **Lemma 2.2.1.2** If

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

is a short exact sequence of modules over R, then

$$length_R(M) = length_R(M') + length_R(M'')$$

- **Proof** Via inequality:
- $length_R(M) \leq length_R(M') + length_R(M'')$: We have the following filtration:For (E^i) of M', (F^i) of M'', and we take the image and inverse image of filtrations:

  $$E^0 \subseteq \cdots \subseteq \cup E^i \subseteq F^0 \subseteq \cdots \subseteq M$$

- $length_R(M) \geq length_R(M') + length + R(M'')$:

$$0 \subseteq E^0 \cap M' \subseteq \cdots \subseteq M' \qquad F^0 \subseteq \cdots \subseteq M''$$

Such $E^0 \subseteq \cdots \subseteq M' \subseteq F^0 \cdots \subseteq M$ is a filtration of M.

□

Local Rings

◇ **Lemma 2.2.1.3** Let (R, m) be a local ring with maximal ideal m. If M is an R-module and $m^n M \neq 0, n \geq 0$.
If M is an R-module and $m^n M \neq 0$ for all $n \geq 0$, then $length_R(M) = \infty$.
In other word: M has finite length if $m^n M = 0$ for some n.

- **Proof** Choose $f_1, \ldots, f_n \in m$, where $f_1 \ldots f_n x \neq 0$ Then, the filtration

$$0 \subseteq Rf_1 \ldots f_n x \subseteq \cdots \subseteq Rx \subseteq M$$

are **distinct**: Prove by contradiction, if $Rf_1 f x = Rf_1 f f_2 x \Rightarrow f_1 x = g f_1 f_2 x \Rightarrow (1 - g f_2) f_1 x = 0 \Rightarrow f_1 x = 0$
Remark: $1 - g f_2 \in R - m$. □

Remark of $m^n M$: Furthermore, in algebra, we can construct the m-**filtration**.

Ring Maps

◇ **Lemma 2.2.1.4** Let $R \to S$ be a ring map. Let M be an S-module. We always have:

$$length_R(M) \geq length_S(M)$$

If $R \to S$ is surjective, then the equality holds.

- **Proof** ... □

Length-Quotient of Fields

◇ **Lemma 2.2.1.5** Let R be a ring with maximal ideal m. Suppose that M is an R-module with

$$mM = 0$$

Then, the length of M as an R agrees with the dimension of M as R/m vector space.

$$length_R(M) = dim_{R/m}(M)$$

The length is finite iff M is a finite R-module.

- **Proof** ... □

Localization

◇ **Lemma 2.2.1.6** Let R be a ring, and M be an R-module.
Let $S \subseteq R$ be a multiplicative subset, then

$$length_R(M) \geq length_{S^{-1}R}(S^{-1}M)$$

∘ **Proof** ... □

Finitely Generated Maximal Ideal

◇ **Lemma 2.2.1.7** ...

∘ **Proof** ... □

2.2.2 Simple Modules

Definition 2.2.2.1 Let R be a ring. M be an R-module. M is **simple** if every submodule of M is either M or 0.

◇ **Lemma 2.2.2.2** Let R be a ring and M be an R-module. The following are equivalent:

- M is simple;
- $length_R(M) = 1$;
- $M \cong R/m$ for some maximal ideal $m \subset R$.

∘ **Proof**

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□

◇ **Lemma 2.2.2.3** Let R be a ring, let M be a finite length R-module. Choose any maximal chain of submodules

$$0 = M_0 \subseteq M_1 \subseteq \cdots \subseteq M_n = M; M_i \neq M_{i-1}$$

∘ **Proof** ... □

◇ **Lemma 2.2.2.4** Let

- A be a local ring with maximal ideal m;
- B is a semi-local ring with maximal ideals $m_i; i = 1, \ldots, n$,

Suppose that $A \to B$ is a homomorphism such that: each m_i lies over m nd such that

$$[\kappa(m_i) : \kappa(m)] < \infty$$

Let M be a B-module of finite length, then

...

In particular, $length_A(M) < \infty$.

∘ **Proof** ... □

2.2.3 Flat Local Homomorphism

◇ **Lemma 2.2.3.1** ...

∘ **Proof** ... □

◇ **Lemma 2.2.3.2** ...

∘ **Proof** ... □

2.3 Artinian Rings, Modules

Definition 2.3.0.1

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- A ring R is **Artinian** if it satisfies the DDC;

Finite Dimensional Algebra

◇ **Lemma 2.3.0.2** Suppose R be is a finite dimensional algebra over a field. Then, R is artinian.

∘ **Proof** ... □

Equivalent Conditions of Artinian Ring

◇ **Lemma 2.3.0.3** ...

∘ **Proof** ... □

2.3.1 Jacobson Radical

◇ **Lemma 2.3.1.1** ...

∘ **Proof** ... □

◇ **Lemma 2.3.1.2** ...

∘ **Proof** ... □

2.4 K-Groups

Definition 2.4.0.1 Let R be a ring. We will introduce the two abelian groups associated to R:

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- $K_0(R)$ with follwing properties
  1. For every finite projective R-module M, there is given an element $[M]$ in $K_0(R)$;
  2. For every short exact sequence
     $$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$
     of finite projective R-module. Then we have $[M] = [M'] + [M'']$.
  3. The group $K_0(R)$ is generated by elements $[M]$;
  4. All relations in $K_0(R)$ are $\mathbb{Z}$-linear combinations of the relations comming from exact sequences as above.

Chapter 3 Integrality

3.1 Integral Closure

Definition 3.1.0.1 Let S be a ring, and $R \subseteq S$ be a subring:

- Let $s \in S$, a monic polynomial

$$g = x^n + a_1 x^{n-1} + \cdots + a_n$$

  with $g(s) = 0$ is called an **integral equation** for s over R.
- An element $s \in S$ is **integral** over R if there exists an integral equation for s over R.
- S is called **integral** over R if every element of S is integral over R.

More generally, we can consider the imbedding $R \hookrightarrow S$ and $s \in S$ is integral if there exists some polynomial $P \in R[X]$ with $P^{\varphi}(s) = 0$.

◇ **Lemma 3.1.0.2** (Nakayama's Lemma)Let $\varphi : R \to S$ be a ring map, and $y \in S$. If there exists a **finite** R-submodule S such that $1 \in M$, $yM \subseteq M$, then y is integral over R

- **Proof** Consider the map

$$M \to M \qquad x \mapsto y \cdot x$$

Then, according to Nakayama's lemma, there exists a polynomial such that $P \in R[X]$, $P^{\varphi}(s) = 0$ □

◇ **Lemma 3.1.0.3** A finite ring map is integral.

- **Proof** Let $S = R$, then the map is finite. □

◇ **Lemma 3.1.0.4** Let $\varphi : R \to S$ be a ring map. Let $s_1, \ldots, s_n$ be a finite set of elements in S.
In this case, s_i is integral over R iff there exists an R-subalgebra $S' \subseteq S$ over R containing all of the s_i.

- **Proof** □

◇ **Lemma 3.1.0.5** Let $R \to S$ be a ring map. The following are equivalent:

- $R \to S$ is finite;
- $R \to S$ is integral and of finite type;
- There exists $x_1, \ldots, x_n$ which generate S as an algebra over R such that each x_i is integral over R.

- **Proof** □

◇ **Lemma 3.1.0.6** Suppose $R \to S$ and $S \to T$ be integral ring maps. Then, $R \to T$ is integral.

◦ **Proof** □

3.1.1 Integral Closure

◇ **Lemma 3.1.1.1** Let $R \to S$ be a ring homomorphism. Then, the collection

$$\bar{S} = \{s \in S | s \text{ is integral over } R\}$$

is an R-subalgebra of S.

◦ **Proof** □

Definition 3.1.1.2

3.2 Normal Rings

Definition 3.2.0.1

3.3 Going Up, Going Down

Suppose p, p' are primes of the ring R. Let $X = Spec(R)$ with the Zariski topology/
Denote $x \in X$ the point corresponding to p, and $x' \in X$ to p'. Then, we have

$$x' \rightsquigarrow x \Leftrightarrow p' \subseteq p$$

In words: x is a specialization of $x' \Leftrightarrow p' \subseteq p$

Definition 3.3.0.1 Let $\varphi : R \to S$ be a ring map:

- We say a $\varphi : R \to S$ satisfies **going up** if given primes $p \subseteq p'$ in R, and a prime q in S lying over p, there exists a prime q' of S such that
  - $q \subseteq q'$;
  - q' lies over p';
- We say a $\varphi : R \to S$ satisfies **going down** if given primes $p \subseteq p'$ in R, and a prime q' in S liying over p', there exists a prime q of S such that
  - $q \subseteq q'$;
  - q lies over p;

Conclusions

- An integral ring map satisfies going up;

  As a special case,
  - Finite ring maps
  - Quotient Maps $R \to R/I$

- Flat ring map satisfies going up;
  As a special case, localization;
-

◇ **Lemma 3.3.0.2** ...

∘ **Proof** ... □

Chapter 4 Algebras I

4.1 The Separable Extension I

4.2 Geometrically Reduced Algebras

4.3 Separable Extensions II

4.4 Perfect Fields

4.5 Univeral Homeomorphisms

4.6 Geometrically Irreducible Algebras

4.7 Geometrically Connected Algebras

4.8 Geometrically Integral Algebras

4.9 Valuation Ring

Given in Valuation-Theory

Chapter 5 Associated Prime,Prime Decompositions

5.1 Supports and Annihilators

Support

Definition 5.1.0.1 Let R be a ring and let M be an R-module. The support of M is the set

$$supp(M) := \{p \in Spec(R) | M_p \neq 0\}$$

◇ **Lemma 5.1.0.2** Let R be a ring, M be an R-module. Then:

$$(M) = 0 \Leftrightarrow Supp(M) = \emptyset$$

∘ **Proof** ⇒ Omitted
⇐ Notice that, for the maximal ideal $m \in Spec(R)$, $M_m = 0 \Rightarrow M = (0)$ □

Annihhilator

Definition 5.1.0.3 Let R be a ring, let M be an R-module.

- Given an element $m \in M$, the annihilator of m is the idea

$$Ann_R(m) := Ann(m) = \{f \in R | fm = 0\}$$

- The annihilator of M is the ideal

$$Ann_R(M) = Ann(M) = \{f \in R | fm = 0, \forall m \in M\} = \cap_{m \in M} Ann_R(m)$$

◇ **Lemma 5.1.0.4** Let $R \to S$ be a flat ring map. Let M be an R-module and $m \in M$.
Then,

$$Ann_R(m)S = Ann_S(m \otimes 1)$$

If M is a finite R-module, then

$$Ann_R(M)S = Ann_S(M \otimes_R S)$$

∘ **Proof** We obtain (2) from (1).
Notice that:

$$Ann_R(m)S = \{f \in R | fm = 0\}S = \{f \otimes s \in R \otimes S | f \in R, fm = 0\} = Ann_S(M \otimes_R S)$$

□

◇ **Lemma 5.1.0.5** Let R be a ring and let M be an R-module.
If M is finite, then $supp(M)$ is closed. More precisely, if $I = Ann(M)$ is the annihilator of M, then

$$V(I) = Supp(M)$$

∘ **Proof** We will show $Supp(M) = V(I)$.
$\subseteq$: Suppose that $p \in Supp(M) \Rightarrow M_p \neq 0 \Rightarrow$
$\supseteq$: □

◇ **Lemma 5.1.0.6** Let $R \to R'$ be a ring map and let M be a finite R-module. Then,

$$Supp(M \otimes_R R')$$

is the inverse image of $Supp(M)$.

∘ **Proof** ... □

◇ **Lemma 5.1.0.7** Let R be a ring, let M be an R-modules, and let $m \in M$.
Then, $p \in V(Ann(m)) \Leftrightarrow m$ does not map to zero in M_p

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∘ **Proof** ... □

◇ **Lemma 5.1.0.8** Let R be a ring and let M be an R-module.
If M is finitely presented R-module, then $Supp(M)$ is a closed subset of $Spec(R)$, whose complement is quasi-compact.

∘ **Proof** Choose a presentation

$$R^{\oplus m} \to R^{\oplus n} \to M \to 0$$

□

◇ **Lemma 5.1.0.9** Let R be a ring and let M be an R-module:

- If M is finite, then the support

$$Supp(M/IM) = Supp(M) \cap V(I)$$

- If $N \subseteq M$, then

$$Supp(N) \subseteq Supp(M)$$

- If Q is a quotient module of M, then $Supp(Q) \subseteq Supp(M)$;
- If $0 \to N \to M \to Q \to 0$ is a short exact sequence, then $Supp(M) = Supp(Q) \cup Supp(N)$.

∘ **Proof** ... □

5.2 Associated Prime

Definition 5.2.0.1 Let R be a ring. Let M be an R-module.

A prime p of R is **associated to** M if there exists an element $m \in M$ whose annihilators is p. i.e. $p = \{x \in R | xm = 0\}$ and p is prime.

Take the following notation of collections

$$Ass_R(M)$$

Remark Not every $Ann(m)$ is prime.

♣ **Example 5.2.0.2**

◇ **Lemma 5.2.0.3** Let R be a ring. M be an R-module. Then

$$Ass(M) \subseteq supp(M)$$

∘ **Proof** □

5.2.1 Associated Prime under Some of Structures

Exact Sequence

◇ **Lemma 5.2.1.1** Let R be a ring. Let $0 \to M' \to M \to M'' \to 0$ be a short exact sequence, then

$$Ass(M) \subseteq Ass(M') \cup Ass(M'')$$

Also, $Ass(M' \oplus M'') = Ass(M') \cup Ass(M'')$.

∘ **Proof** It is easy to check the inclusion

$$Ass(M') \subseteq Ass(M)$$

□

Filtrations

◇ **Lemma 5.2.1.2** Let R be a ring, M an R-module. Suppose there exists a filtration by R-submodules:

$$0 = M_0 \subseteq M_1 \subseteq \cdots \subseteq M_n = M$$

such that

∘ **Proof** ... □

Associated Prime of Finite Module over Noetherian Rings

◇ **Lemma 5.2.1.3** ...

∘ **Proof** ... □

Minimal Primes

♠ **Proposition 5.2.1.4** Let R be a Noetherian ring. Let M be a finite R-module. The following sets of primes are the same:

- The minimal primes in the support of M;
- The minimal primes in $Ass(M)$;
- For any filtration

  $$0 = M_0 \subseteq \cdots \subseteq M_n = M \qquad M_i/M_{i-1} \cong R/p_i$$

  the minimal primes of the set $\{p_i\}$.

◦ **Proof**

- (1)⇔ (3) By the conclusion:
-
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□

Chapter 6 Spectra

6.1 Open and Closed Subsets

Definition 6.1.0.1 Define:

◇ **Lemma 6.1.0.2** ...

∘ **Proof** ... □

6.2 Connected Components

6.3 Glueing Functions

Chapter 7 Dimension

7.1 Dimension of Rings

7.1.1 Definitions: Chain, Length, Superemum, Height

Definition 7.1.1.1

- Let R be a ring. A **chain of prime ideals** is a sequence

$$p_0 \subseteq p_1 \subseteq \cdots \subseteq p_n$$

  of prime ideals such that $p_i \neq 0 p_{i+1}$;

- Define:

$$length((p_i)) := n$$

- (Krull Dimension) Let R be a ring. Then, the **superemum of integers** $n \geq 0$ such that R has a chain of prime ideals

$$p_0 \subseteq p_1 \subseteq \cdots \subseteq p_n; p_i \neq p_{i+1}$$

  of length n. i.e. $dim(R) = \sup(length(p_i))$;

- (Height) Define:

$$ht(p) := dim(R_p)$$

◇ **Lemma 7.1.1.2** The **Krull dimension** of R is the superemum of the heights of its (maximal) primes.

- **Proof** $dim(R) := \sup_{(p_i)} length(p_i)$, the superemum must be a chain to maximal ideal $m \in Spec_{max}(R)$. Thus

$$dim(R) = \sup_{(p_i), end(p_i) \in Spec_{max}(R)} length(p_i)$$

□

◇ **Lemma 7.1.1.3** A Noetherian ring of dimension 0 is Artinain.
Conversely, any artinain ring is Noetherian of dimension zero.

- **Proof**

□

◇ **Lemma 7.1.1.4** Let R be a Noetherian local ring. Then $dim(R) = 0 \Leftrightarrow d(R) = 0$. **Remark** Notion $d(-)$: Here

- **Proof** ... □

♠ **Proposition 7.1.1.5** Let R be a ring. The following are equivalent:

- R is Artinian;

- R is Noetherian and $dim(R) = 0$;
- R has finite length as a module over itself;
- R is a finite product of Artinain local rings;
- R is Noetherian and $Spec(R)$ is a finite discrete topological space;
- R is a finite product of Noetherian local rings of dimension 0;
- R is a finite product of Noetherian local rings R_i with $d(R_i) = 0$;
- R is a finite product of Noetherian local rings R_i whose maximal ideals are nilpotent
- R is Noetherian, has finitely many maximal ideals and its Jacobson radical ideal is nilpotent;
- R is Noetherian and no strict inclusions among its primes;

∘ **Proof** ... □

◇ **Lemma 7.1.1.6** Let R be a local Noetherian ring, the following are equivalent:

- $dim(R) = 1$;
- $d(R) = 1$;
- there exists an $x \in m$, x not nilpotent such that $V(x) = \{m\}$;
- there exists an $x \in m$, x not nilpotent such that $m = \sqrt{(x)}$;
- there exists an ideal of definition generated by 1 element, and no ideal of definition is generated by 0 elements.

∘ **Proof** ... □

♠ **Proposition 7.1.1.7** Let R be a local Noetherian ring, $d \geq 0$ be an integer, the followings are equivalent:

- $dim(R) = d$;
- $d(R) = d$;
- there exists an ideal of definition generated by d eleemnts, and no ideal of definition is generated by fewer than d elements;

∘ **Proof** ... □

7.1.2 Noetherian Local Ring

Definition 7.1.2.1 Let (R, m) be a local ring. Define:

$$dim_{emb}(R) := dim_{\kappa(m)} m/m^2$$

◇ **Lemma 7.1.2.2** Let (R, m) be a Noetherian local ring, then

$$dim(R) \leq dim_{\kappa(m)} m/m^2$$

∘ **Proof** Via Nakayama's lemma, suppose $x_1, \dots, x_n$ generates $m/m^2 = M/IM$, $M = m$, $I = m$, and $m \subseteq rad(R) = \cap_{m' \in Spec_{max}(R)} m' \Rightarrow x_1, \dots, x_n$ generates m, and $n = dim_{\kappa(m)} \geq dim(R)$ □

Definition 7.1.2.3 Let R be a Noetherian ring, $x \in R$:

- A **system of parameters** of R is a sequence of elements $x_1, \dots, x_d \in m$ which generates an ideal of definition of R; 9.2.1.1
- If there exists $x_1, \dots, x_d \in m$ such that

  $$m = (x_1, \dots, x_d)$$

  then we call R a **regular local ring** and $x_1, \dots, x_d$ a **regular system of parameters**.

◇ **Lemma 7.1.2.4** Let (R, m) be a Noetherian local ring of dimension d

-
-

∘ **Proof** ... □

7.2 Application of Dimension Theory

◇ **Lemma 7.2.0.1** Let R be a Noetherian local domain of dimension ≥ 2. A nonempty open subset $U \subseteq Spec(R)$ is infinite.

∘ **Proof** ... □

◇ **Lemma 7.2.0.2** A Noetherian ring with finitely many primes has dimension ≤ 1.

∘ **Proof** ... □

◇ **Lemma 7.2.0.3** Let S be a nonzero finite type algebra over a field k. The following are equivalent:

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-
-
-
-
-

-
-

∘ **Proof** ... □

7.2.1 Noetherian Jacobson Rings

◇ **Lemma 7.2.1.1** (Noetherian Jacobson Rings)

- Any Noetherian domain R of dimension 1 with finitely many primes is Jacobson;
- Any Noetherian ring such that every prime p is either maximal or contained in infinitely many prime ideals is Jacobson.

∘ **Proof** ... □

7.3 Support, and Dimension of Modules

◇ **Lemma 7.3.0.1** Let R be a Noetherian ring, and M be a finite R-module, there exists a filtration by R-modules

$$0 = M_0 \subseteq M_1 \subseteq \cdots \subseteq M_n = M$$

such that each quotient $M_i/M_{i-1} \cong R/p_i$ for some prime ideal p_i of R.

∘ **Proof** ... □

Chapter 8 Associated Primes

8.1 Associated Primes

Definition 8.1.0.1 Let R be a ring, M be an R-module.

A prime p of R is associated to M if there exists an element $m \in M$ whose annihilator is p. The set of all such primes is denoted

$$Ass_R(M) \qquad \text{or} \qquad Ass(M) = \{p \in Spec(R) | \exists m \in M, p = Ann(m)\}$$

Remark Not every annihilator of principle element is prime. For example:

...

◇ **Lemma 8.1.0.2** Let R be a ring, M be an R-module, then

$$Ass(M) \subseteq Supp(M)$$

∘ **Proof** ... □

8.2 Symbolic Powers

8.3 Relative Assassin

8.4 Weakly Associated Primes

8.5 Embedded Primes

Chapter 9 Hilbert-Sammuel Polynomials

9.1 Polynomials

9.1.1 Integral-Valued Polynomials

Definition 9.1.1.1 A **binomial polynomial** $k = 0, 1, \dots$ are:

- $Q_0(X) := 1$;
- $Q_1(X) := X$;
- $Q_k(X) := \binom{X}{k}$;

And **difference operator**

$$\Delta f(n) := f(n+1) - f(n)$$

Then:

$$\binom{X}{k+1} - \binom{X}{k} = \binom{X}{k-1} \Rightarrow \Delta Q_k = Q_{k-1}$$

♠ **Proposition 9.1.1.2** Let $f \in \mathbb{Q}[X]$ be a polynomial. The following properties are equivalent:

- f is $\mathbb{Z}$-linear combination of binomial polynomial Q_k;
- For all $n \in \mathbb{Z}$, $f(n) \in \mathbb{Z}$;
- $f(n) \in \mathbb{Z}$ for n large enough;
- Δf is $\mathbb{Z}$-linear combination of binomial polynomials.

∘ **Proof** Clearly $(a) \Leftrightarrow (b)$; $(a) \Leftrightarrow (d)$.

- $(a) \Leftrightarrow (c) \Leftarrow$ We use induction on degree of f induced by $deg(f) = deg(P) - deg(Q), f = P/Q$.
  $deg(f) = 0$ Omitted;
  Induction Steps: Apply (4).

□

Definition 9.1.1.3 If f satisfies one of the structure above, then f is called **integral-valued**. With following decomposition

$$f = \sum c_k Q_k$$

9.1.2 Polynomial-Like Functions

Definition 9.1.2.1 $f : \mathbb{Z} \to \mathbb{Z}$ is a **polynomial-like** function if there exists some polynomial $P_f(X) \in \mathbb{Z}(X)$ such that

$$P_f(n) = f(n); n >> 0$$

♠ **Proposition 9.1.2.2** The following conditions are equivalent:

- f is polynomial-like;
- Δf is polynomnial linke;
- There exists $r \geq 0$ such that $\Delta^r f(n) = 0$ for all large enough n.

∘ **Proof** (1)⇔ (2) Omitted;
(1)⇔ (3)

- $\Rightarrow r = deg(f)$;
- $\Leftarrow$ Integral r times.

□

A (small?) Approxiamation Problem

Exists $f \sim Q$? $f \sim P_f, Q \sim P_f \Rightarrow f \sim Q \Rightarrow Q \in \mathbb{Z}[X]$ failed.

9.1.3 The Hilbert Polynomials

title

Definition 9.1.3.1 Consider a graded ring $H = \oplus H_n$ having the following properties:

- H_0 is Artinian;
- The ring H is generated by H_0 and by a finite number $x_1, \ldots, x_r \in H_1$.

Thus, H is the quotient of polynommial ring $H_0[X_1, \ldots, X_r]$, with function

$$\chi(M, n) := l_{H_0}(M_n)$$

We have
♠ **Proposition 9.1.3.2** ...

∘ **Proof** ... □

▶ ***Theorem*** (Hilbert) $\chi(M, n)$ is polynomial-like function.

∘ **Proof** ... □

▶ ***Theorem*** Assume M_0 generates M as H-module, then:

- $\Delta^{r-1}Q(M) \leq l(M_0)$;
- The following properties are equivalent:
  - $\Delta^{r-1}Q(M) = l(M_0)$
  - $\chi(M, n) = l(M_0)\binom{n+r-1}{r_1}, n \geq 0$

- The natural map $M_0 \otimes_{H_0} H_0[X_1, \dots, X_r] \to M$ is an isomorphism.

•

○ **Proof** ... □

title

Definition 9.1.3.3 Let (R, m, κ) be a Noetherian local ring. Let M be a finite R-module, we define

- **Hilbert function** of M to be function

$$\varphi_M : n \mapsto lenght_R(m^n M / m^{n+1} M)$$

-

$$\chi_M : n \mapsto length_R(M / m^{n+1} M)$$

◇ **Lemma 9.1.3.4** It is easy to verify

$$\chi_M(n) = \sum_{i=0}^{n} \varphi_M(i)$$

○ **Proof** By property of length. □

9.1.4 The Samuel Polynomials

◇ **Lemma 9.1.4.1** Let A be a noetherian commutative ring, M be finitely generated A-modules, q is an ideal of A, then,

$$l(M/qM) < \infty \Leftrightarrow Supp(M) \cap V(q) \text{ are maximal ideals}$$

○ **Proof** ... □

★ **Corollary 9.1.4.2** ...

○ **Proof** ... □

Definition 9.1.4.3 ...

♠ **Proposition 9.1.4.4** ...

○ **Proof** ... □

Remark Psychology of Intuition of Hilbert's polynomial:

9.2 Noetherian Local Ring

Definition 9.2.0.1 Hilbert functions of module: Define

$$\varphi_M : n \mapsto length(m^n M / m^{n+1} M)$$

9.2.1 Ideal of Definition of R

Definition 9.2.1.1 Let (R, m) be local Noeterian ring. An ideal $I \subseteq R$ such that

$$\sqrt{I} = m$$

9.2.2 Hilbert Polynomial of M over R

Definition 9.2.2.1 ...

9.3 Poincare Series

Chapter 10 Multiplicities

10.1 Multiplicities of a Ring

10.2 Intersection Multiplicities of Two Modules

10.3 Tor-Formula

Chapter 11 Graded Rings

11.1 Graded Rings,Graded Module

11.1.1 Definitions

Definition 11.1.1.1

- Let S be a ring, endowed with a direct sum composition

$$S = \oplus_{d \geq 0} S_d$$

  which satisfies $S_d \cdot S_e \subseteq S_{d+e}$.
- With this graded ring, define graded module

$$M = \oplus_{d \geq 0} M_d$$

  which satisfeis $S_d \cdot M_e \subseteq M_{d+e}$.

Remark We set $S_{-k} = 0, k \geq 0$.

- Especially, the **irrelevant ideal** is the ideal

$$S_+ = \oplus_{d>0} S_d$$

Homogeneous Elements

Definition 11.1.1.2

- An element $x \in S$ is called **homogenous** if $x \in S_d$ for some $d \geq 0$;
- An element $x \in M$ is called **homogenous** if $x \in M_d$ for some $d \geq 0$;

Graded Ring/Module Homomorphsim

Definition 11.1.1.3

- In general, let $M = \oplus M_d, M' = \oplus M'_d$ be two S-graded modules. The **S-graded map** is

$$\varphi : M \to M'$$

  such that $\varphi(M_d) \subseteq M'_d$. Take the following notation:

$$GrHom_0(M, N) = \{f : M \to N | f \text{ is graded}\}$$

  which is an S_0-module. (The multiplication of the elements in S_0 does not change the homogeneous degree)
- Take the following notation

$$M(n)_d := M_{n+d}$$

  which is called **twist** of M by n.

$$GrHom(M, N) := \oplus_{n \in \mathbb{Z}} GrHom_n(M, N) \qquad GrHom_n(M, N) = GrHom_0(M, N(n))$$

Especially, there exists natural isomorphism

$$(M \oplus N)(n) = M(n) \oplus N(n) \qquad (M \otimes_S N)(n) = M(n) \otimes_S N(n)$$

Sub-Graded Ring/Module,Quotients

As what we did for ring and module. Omitted.

11.1.2 Properties

◇ **Lemma 11.1.2.1** (Nakayama) Let S be a graded ring and M a graded S-module:

- $S^{+}M = M$, M is finite$\Rightarrow M = 0$;
- $N, N' \subseteq M$ are graded submodule, $M = N + S_{+}N'$, N' is finite, then $M = N$;
- If $N \to M$ is a map of graded modules,

  $$N/S_{+}N \to M/S_{+}M$$

  is surjective, M is finite, $N \to M$ is surjective;
- $x_1, \dots, x_n \in M$ are homogeneous,generate $M/S_{+}M$ and M is finite. Then $x_1, \dots, x_n$ generates M.

∘ **Proof**

- By induction on the number of generators r: Choose the generators $y_1, \dots, y_r$ where y_i is homogeneous, $deg(y_i) = d_i$. Now select $d_r = min(d_i)$ by rearrangement Thus $y_r = 0$ by $d_r \geq d_r + 1$ if $y_r \neq 0$.
- Apply (1). $M/N = S_{+}N'/S_{+}N' \cap N \subseteq S_{+}(N'/N) \subseteq S_{+}(M/N)$ where $N'/N \subseteq M/N$ as a submodule. Thus we have $M/N = 0 \Rightarrow M = N$;
- Consider $M = Im(N \to M) + S_{+}0$ so $M = Im(N \to M)$.
- Consider the morphism

  $$\oplus S(-d_i) \to M \qquad (s_1, \dots, s_n) \mapsto \sum s_i x_i$$

□

Module $S^{(d)}$

Definition 11.1.2.2 Take the following notation

$$(S^d_n) := S_{nd} \qquad (M^d)_n := M_{nd}$$
$$S^{(d)} := \oplus_{n\geq 0} S_{nd} \qquad M^{(d)} := \oplus_{n\geq 0} M_{nd}$$

◇ **Lemma 11.1.2.3** Let S be a graded ring, which is finitely generated over S_0. Then for all sufficiently divisable d the algebra $S^{(d)}$ is generated in degree 1 over S_0.

∘ **Proof**

□

Integral Closure

◇ **Lemma 11.1.2.4** Let $R \to S$ be a homomorphism of graded rings. $S' \subseteq S$ be the integral closure of R in S. Then

$$S' = \oplus_{d \geq 0} S' \cap S_d$$

i.e. S' is a graded R-subalgebra.

- **Proof**

□

Graded Free

Definition 11.1.2.5 A graded S-module M is called **graded-free** if it has a Basis made up of homogeneous elements. i.e. Homogeneous basis B with $S(B) = M$.

◇ **Lemma 11.1.2.6** (Serre, Local Algebra, P92)The followings are equivalent:

- M is graded-free;
- M is S-flat as a nongraded module;
- $Tor_1^A(M, k) = 0$;

∘ **Proof** ... □

11.2 Proj Constructions

11.2.1 Graded Ideal

Definition 11.2.1.1 Let S be a graded ring. A **homogeneous ideal** is simly an ideal $I \subseteq S$ which is also a graded submodule of S.

Equivalently, it is an ideal generated by homogeneous elements.

Equivalently, if $f \in I$, f has a unique decomposition

$$f = f_0 + \cdots + f_n; f_i \in S_i \cap I$$

Notation Take the following notation for **homogeneous graded ideals**: Suppose S be

$$Spec^h(S) := \{p \in Spec(S) | p \text{ is graded ideal}\}$$

11.2.2 Proj Construction

Definition 11.2.2.1 Let S be a graded ring. We define:

$$Proj(S) := \{p \in Spec^h(S) | S_+ \nsubseteq p\}$$

And $Proj(S)$ is equipped with the Zariski subspace topology, as a subcollection of $Spec(S)$. It is called the **homogeneous spectrum** of the graded ring S.

It is easy to verify: $Proj(S)$ is a topological subspace.
The space is equipped with a continuous map via inclusion

$$Proj(S) \to Spec(S_0) \qquad S_0 \hookrightarrow S$$

Definition 11.2.2.2 Define the **graded localization**

- Suppose S be a multiplicative homogeneous subset. Then, define:

$$(S^{-1}R)_n := \{\frac{r}{s}|r \in R_k, s \in S \cap R_{k-n}\}$$

- Especially, let $f \in S$, define:

$$S_{(f)} := \{\frac{r}{f^n}|r \in S, deg(r) = nd\}, d = deg(f)$$
$$M_{(f)} := \{\frac{r}{f^n}|deg(r) = nd, r \in M, d = deg(f)\}$$

◇ **Lemma 11.2.2.3** Let S be a $\mathbb{Z}$-graded ring containing a homogeneous invertible element of positive degree. Then,

$$G = \{\mathbb{Z}\text{-graded primes of } S\} \to Spec(S_0)$$

is a homeomorphism.

∘ **Proof** □

Definition 11.2.2.4

- For $f \in S$ homogeneous of degree > 0, define:

$$D_+(f) := \{p \in Proj(S)|f \notin p\}$$

- For homogeneous ideal $I \subseteq S$, define

$$V_+(I) := \{p \in Proj(S)|I \subset p\}$$

Especially: For set $E \subseteq S$, take notion $V_+(E) := V_+((E))$.

◇ **Lemma 11.2.2.5** (Topology of Proj) Let $S = \oplus_{d\geq 0} S_d$

- $D_+(f)$ is open in $Proj(S)$;
- $D_+(ff') = D_+(f) \cap D_+(f')$;
- Let $g = g_0 + \cdots + g_m$ be an element of S with $g_i \in S_i$,

$$D(g) \cap Proj(S) = (D(g_0) \cap Proj(S)) \cup \bigcup_{i\geq 1} D(g_i)$$

- Let $g_0 \in S_0$ be a homogeneous element of degree 0, then

$$D(g_0) \cap Proj(S) = \cup_{f\in S_d, d\geq 1} D_+(g_0 f)$$

- The open set $D_+(f)$ form a basis for the topology of $Proj(S)$;

- Let $f \in S$ be homogeneous of positive degree. The ring S_f is a natural $\mathbb{Z}$-grading. The ring maps

$$S \to S_f \leftarrow S_{(f)}$$

induce homeomorphisms:

$$D_+(f) \leftarrow \{\mathbb{Z}\text{-graded primes of } S_f\} \to Spec(S_{(f)})$$

- There exists an S such that $Proj(S)$ is not quasi-compact;
- The sets $V_+(I)$ is closed;
- Any closed subset $T \subseteq Proj(S)$ is of the form $V_+(I)$ for some homogeneous ideal $I \subseteq S$;
- For any graded ideal $I \subseteq S$ we have $V_+(I) = \emptyset$ iff $S_+ \subseteq \sqrt{I}$;

∘ **Proof** □

♣ **Example 11.2.2.6** Let R be a ring, $S = R[X]$ with $deg(X) = 1$. Then, the map

$$Proj(S) \to Spec(R)$$

is a bijection in fact a homeomorphism. Namely, suppose $p \in Proj(S)$.

◇ **Lemma 11.2.2.7** Let S be a graded ring. M be a graded S-module and $p \in Proj(S)$.
Let $f \in S$ be a homogneous element of positive degree such that $f \notin p$ i.e. $p \in D_+(f)$.
Let $p' \subseteq S_{(f)}$ be the element of $Spec(S_{(f)})$ corresponding to p as lemma above, then

$$S_{(p)} = (S_{(f)})_{p'} \qquad M_{(p)} = (M_{(f)})_{p'}$$

∘ **Proof** ... □

◇ **Lemma 11.2.2.8** ...

∘ **Proof** ... □

Localization

◇ **Lemma 11.2.2.9** Let S be a graded ring, M be a graded S-module. Let p be an element of $Proj(S)$, let $f \in S$ be a homogeneous element of positive degree such that

$$f \notin p \text{ or } p \in D_+(f)$$

Let $p' \subseteq S_{(f)}$ be the element of $Spec(S_{(f)})$ corresponding to p

---

∘ **Proof** ... □

Prime Avoidance

Finite Type Algebra

11.3 Noetherian Graded Rings

11.3.1 Noetherian Graded Ring Criterion

◇ **Lemma 11.3.1.1** Let S be a graded ring. A set of homogeneous elements $f_i \in S_+$ generates S as an algbra over S_0 iff they generate S_+ as an ideal of S.

∘ **Proof** ... □

◇ **Lemma 11.3.1.2** A graded ring S is Noetherian iff S_0 is Noetherian and S_+ is finitely generated as an ideal of S.

∘ **Proof**

- $\Rightarrow$
- $\Leftarrow$

□

11.3.2 Numerical Polynomial

Definition 11.3.2.1 Let A be an abelian group. We say that a function $f : n \mapsto f(n) \in A$ defined for all sufficient large integers n is a **numberical polynomial** if there exists $r \geq 0$, elements ${}_0, \dots, a_r \in A$ such that

$$f(n) = \sum_{i=1}^{r} \binom{n}{i} a_i, n >> 0$$

◇ **Lemma 11.3.2.2** ...

∘ **Proof** ... □

◇ **Lemma 11.3.2.3** Suppose $f : n \mapsto f(n) \in A$ defined for all n sufficiently large and suppose that

$$n \mapsto f(n) - f(n-1)$$

is a nummerical polynomial, then f is a numerical polynomial.

∘ **Proof** ... □

◇ **Lemma 11.3.2.4** If M is a finitely generated graded S-module, and if S is finitely generated over S_0, then each M_n is finite S_0-module.

∘ **Proof** ... □

♠ **Proposition 11.3.2.5** Suppose S be a Noetherian graded ring and M a finite graded S-module. Consider the following function

$$\mathbb{Z} \to K_0'(S_0) \qquad n \mapsto [M_n]$$

If S_+ is generated by elements generated by elements of degree 1, then this function is a numerical polynomial.

∘ **Proof** Apply induction on minimal generators of S_1: □

Remark

♣ **Example 11.3.2.6** (Hilbert Polynomial for Syzygies)

◇ **Lemma 11.3.2.7** ...

∘ **Proof** ... □

11.3.3 Noetherian Local Rings

Look here: 9.2

Chapter 12 Blow-Up Algebras/Rees Algebras

Reference: Vasconcelos Integral Closure: Rees Algebras, Multiplicities, Algorithms

Definition 12.0.0.1 Let R be a ring, $I \subseteq R$ be an ideal:

- The **blow up algebra**, or **Rees algebra** associated to the pair (R, I) is the graded R-algebra
- Let $a \in I$ be an element. Denote

Chapter 13 Polynomials

13.1 Noether's Normalization Theorem

▶ ***Theorem*** (Noether's Normalization Theorem) Let k be an infinite field, and

$$A = K[a_1, \dots, a_n]$$

is a finitely generated algebra. Then, there exists a finite $y_1, \dots, y_m, m \leq n$.

- $y_1, \dots, y_m$ are algebraically independent;
- A is a finite $k[y_1, \dots, y_m]$-algebra.

◦ **Proof** □

13.2 Hilbert's Basis Theorem

▶ ***Theorem*** (Hilbert) Suppose that R be a Noetherian ring, so is $R[X]$.

◦ **Proof** Let $I \subseteq R[X]$ be an arbitrary ideal. We need to show: I is finitely generated. Define the collection

$$J_i = \{a_i \in R | \exists a_0, \dots, a_{i-1} \in R, \sum_{k=0}^{i} a_k X^k \in I\}$$

Then, J_i is an R-ideal:
For $\alpha, \beta \in J_i, \gamma \in R$, then they have correspondent polynomials $f, g \in I$. And

$$f + g \in I \Rightarrow \alpha + \beta \in J_i$$
$$\gamma g \in I \Rightarrow \gamma \beta \in J_i$$

J_i are an ascending chain ideals: $\alpha \in J_i$ with $f \in I, Lead(f) = \alpha$. Then $fX \in I \Rightarrow \alpha \in J_{i+1}$. Thus,

$$J_i = (a_{n,1}, \dots, a_{n,r})_R$$

Let r be the maximal number of genrating elements of J_i from 0 to n. Suppose $a_{n,1}, \dots, a_{n_r}$ are correspondent to the polynomials $f_{n,1}, \dots, f_{n,r} \in R[X]$, then define

$$I' := (f_{i,j} | i = 0, \dots, n; j = 1, \dots, r) \subseteq I$$

We will prove: $I' = I$. Suppose $g = \sum b_i X^i \in I$, with $deg(g)$. Consider the degree of g: Apply induction

- $deg(g) < n$: $b_d \in J_d$, there exists representation

$$b_d = \sum_{i=1}^{r} r_i a_{d,i}$$

Then,

$$\hat{f} = f - \sum_{i=1}^{r} r_i f_{d,i} \in I \Rightarrow f \in I$$

- $g \geq n$: Similarly.

Thus, $I = I'$. □

★ **Corollary 13.2.0.1** Suppose R be a Noetherian ring, so is $R[X_1, \dots, X_n]$.

∘ **Proof** Notice,

$$R[X_1, \dots, X_n] = R[X_1, \dots, X_{n-1}][X_n]$$

□

★ **Corollary 13.2.0.2** Suppose R be a Noetherian ring, then any of finitely algebra over R is Noetherian. In particular, every **affine algebra** is Noetherian.

∘ **Proof** Consider the quotient. □

◇ **Lemma 13.2.0.3** Let K be a field, then $K[X]$ is Noetherian.

∘ **Proof** K can be seen as ring with only finite chain of length 1. □

★ **Corollary 13.2.0.4** The polynomial ring $K[X_1, \dots, X_n]$ is Noetherian.

∘ **Proof** As above. □

Remark This is a fundamental result, that the space A^n_K is Noetherian.

13.3 Hilbert's Nullstellensatz

◇ **Lemma 13.3.0.1**

- If A is an integral domain and algebraic over K, then A is a field;
- Let k be a field and K a field extension which is finitely generated as a k-algebra, then K is algebraic over k. Equivalently, for a field A, which is contained in a k-affine domain, then A is algebraic.

∘ **Proof**

-
-

□

▶ ***Theorem***

∘ **Proof** □

▶ ***Theorem***

∘ **Proof** □

◇ **Lemma 13.3.0.2**

∘ **Proof** □

13.3.1 Hilbert's Nullstellensatz, and Applications

▶ ***Theorem*** (Hilbert) Suppose that

∘ **Proof** □

13.3.2 Textual Research: The Original Version by Hilbert

Chapter 14 Compeletion

14.1 Completion

Definition 14.1.0.1 Suppose R be a ring and I be an ideal. We define

- the **compleltion** of R with respect to I to be the limit
-

14.2 Compeletion for Noetherian Rings

14.3 Taking Limits of Modules

Chapter 15 Differentials and Smoothness

15.1 Differentials

15.1.1 Derivative

Definition 15.1.1.1 Suppose $\varphi : R \to S$ be a ring map(thus, S is an R-module) , and M be an S-module. An R-derivation is a map

$$D : S \to M$$

such that

- D is additive: $D(a+b) = D(a) + D(b)$;
- Annihilates elements of $\varphi(R)$:

$$D(\varphi(r)) = 0, r \in R$$

- Lebnitz's Rule:

$$D(ab) = aD(b) + bD(a)$$

Take the following notion of $Der_R(S, M)$ for such $D \in Hom_S(S, M)$.

And notice that,

$$D(ra) = D(\varphi(r)a) = \varphi(r) \cdot D(a)$$

Thus, an equivalent definition is:

- $D : S \to M$ is R-linear;
- D satisfies Lebnitz's rule.

◇ **Lemma 15.1.1.2** $Der_R(S, M)$ is an S-submodule of $Hom_S(S, M)$
Especially, for S-module map $\alpha : M \to N$, consider the composition $\alpha \in D \in Der_R(S, N)$. Thus, $M \mapsto Der_R(S, M)$ is covariant.

∘ **Proof** ... □

To have a deeper undertand of the conception, we could learn elementary knowledge about Lie group/Manifolds.

15.1.2 Kahler Differentials

Definition 15.1.2.1 Consider the following map

$$\bigoplus_{(a,b)\in S^2} S([a,b]) \oplus \bigoplus_{(f,g)\in S^2} S([f,g]) \oplus \bigoplus_{r\in R} S[r] \to \oplus_{a\in S} S[a]$$

$$[(a,b)] \mapsto [a+b] - [a] - [b]$$

$$[(f,g)] \mapsto [fg] - g[f] - f[g]$$

$$[r] \mapsto [\varphi(r)]$$

And then, define $Coker(\bigoplus_{(a,b)\in S^2} S([a,b]) \oplus \bigoplus_{(f,g)\in S^2} S([f,g]) \oplus \bigoplus_{r\in R} S[r] \to \oplus_{a\in S} S[a])$, and a morphism

$$d : S \to Coker$$

Such that $d(a) = \overline{[a]} \in Coker$ which implies

$$d(a+b) = da + db \qquad d(fg) = fdg + gdf \qquad d(\varphi(r)) = 0; a, b, f, g \in S, r \in R$$

which is element of$Der_R(S, \Omega_{S/R} = coker)$
The pair $(\Omega_{R/S}, d)$ is called **the module of Kahler differentials** or the **module of differentials** of S over R.

◇ **Lemma 15.1.2.2** The module of differentials of S over R has the following universal property: The map

$$Hom_S(\Omega_{S/R}, M) \to Der_R(S, M)$$

is an isomorphism of functors.

∘ **Proof** ... □

This is an another consturction of $\Omega_{S/R}$.

♠ **Proposition 15.1.2.3** Especially, if $R \to S$ is surjective, then $\Omega_{S/R} = 0$.

∘ **Proof** ... □

★ **Corollary 15.1.2.4** ...

∘ **Proof** ... □

Universal Property

♠ **Proposition 15.1.2.5** Given $f \in Der_R(S, M)$, there exists factorization

$$\begin{tikzcd} \Omega^1_{B/A} \arrow[r] & M \\ S \arrow[u, dashed, "\exists!"] \arrow[ur] & \end{tikzcd}$$

∘ **Proof** ... □

Kahler Differentials-Zariski Construction

Definition 15.1.2.6 Equivalently: Consider the kernel of

$$I := Ker(S \otimes_R S \to S)$$

Then,

$$\Omega_{S/R} \cong I/I^2$$

With differential map

$$S \to \Omega_{S/R} \qquad a \mapsto [1 \otimes a - a \otimes 1]$$

Now verify the equivalence between the two definitions:

- $I/I^2 \to Coker$: Define the following map

$$\beta : I \to Coker \qquad \sum s_i \otimes t_i \mapsto \sum s_i dt_i$$

Then, for elements in I^2, take

$$\begin{cases} \omega_1 = 1 \otimes a - a \otimes 1 \\ \omega_1 = 1 \otimes b - b \otimes 1 \end{cases}$$

We have

$$\begin{aligned} \beta(\omega_1\omega_2) &= \beta((1 \otimes a - a \otimes 1)(1 \otimes b - b \otimes 1)) \\ &= \beta(1 \otimes (ab) - b \otimes a - a \otimes b + ab \otimes 1) \\ &= d(ab) - bda - adb + abd(1) \\ &= 0 \end{aligned}$$

- $Coker \to I/I^2$: $d : S \to \Omega_{S/R}$ is a derivative. With:
  -
  -

Limits

◇ **Lemma 15.1.2.7** Let I be a directed set, a system of $(R_i \to S_i, \varphi_{ii'})$ be a system of ring maps over$I$, then we have

$$\Omega_{S/R} = colim_i \Omega_{S_i/R_i}$$

where $R \to S = colim(R_i \to S_i)$.

- **Proof** ... □

◇ **Lemma 15.1.2.8**

- **Proof** ... □

Sequence

♠ **Proposition 15.1.2.9** Consider ring map $A \to B \to C$, there exists a canonical exact sequence:

$$...$$

of C-modules.

- **Proof** ... □

Localization,Quotients

title

15.2 De Rham Complex

Recall:

$$\Omega^i_{B/A} := \wedge^i_B(\Omega_{B/A}); i \geq 0$$

Then:

Definition 15.2.0.1 The **de Rham complex** of B over A is the complex of $((\Omega^i_{B/A}), d)$ with the chain map $d : \Omega^i_{B/A} \to \Omega^{i+1}_{B/A}$ constructed as following:

$$d : \Omega^p \to \Omega^{p+1} \qquad f_0 df_1 \wedge \cdots \wedge df_p \mapsto df_0 \wedge df_1 \wedge \cdots \wedge df_p$$

Verify the chain condition

- **Proof**

$$\Omega^0_{B/A} \longrightarrow \Omega^1_{B/A} \longrightarrow \Omega^2_{B/A} \longrightarrow \dots$$

$$\Omega^0 \longrightarrow \Omega^1 \longrightarrow \Omega^2 \longrightarrow \dots$$

□

15.3 Finite Order Differential Operators

Definition 15.3.0.1 Let $R \to S$ be a ring map, M, N be S-modules.Let $k \geq 0$ be an integer.
We inductively define a **differential operator**

$$D : M \to N$$

of order k to be an R-linear map such that $m \mapsto D(gm) - gD(m)$ is a differential operator of order $k-1$ for <u>all</u> $g \in S$.
Especially: The differential operator of order 0 is an S-linear map.
Denote as $Diff^k(M, N)$.

Verify the properties:

- By induction,$g \in S, gD(fm) - fgD(m)$ is differential operator of order $k-1$;
- Apply induction, $(D + D')(gm) - g(D + D')(m)$ is differential operator of order $k-1$;

$$Diff^0(M, N) \subseteq Diff^1(M, N) \subseteq \dots$$

Take the following notion

$$Diff^k_{S/R}(M, N)$$

♠ **Proposition 15.3.0.2** Let $R \to S$ a ring map, L, M, N be S-modules.

If $D : L \to M$ and $D' : M \to N$ are differential operators of order k and k', then $D' \circ D$ is a differential operator of order $k + k'$.

∘ **Proof** ... □

♠ **Proposition 15.3.0.3** Let $R \to S$ be a map. Let M bean S-module. Then, there exixsts an S-module $P^k_{S/R}(M)$ and a canonical isomorphism

$$Diff^k_{S/R}(M, N) = Hom_S(P^k_{S/R}(M), N)$$

Functorial in the S-module N.

Via Yoneda's lemma, this tell us: A 'universal'differential operator

$$D_{univ} : M \to P^k_{S/R}(M)$$

such that: For each $D : M \to N$, which is a , there exists a unique factorization $\alpha : P^k_{S/R}(M) \to N$ such that

$$D = \alpha \circ D_{univ}$$

∘ **Proof** ... □

15.3.1 Modules of Principe Part

Definition 15.3.1.1 Such $P^k_{S/R}(M)$ is called the **sequence of principal parts** of order k of M, or sometimes **Jet module**.

♣ **Example 15.3.1.2** ...

Exact Sequence induced by $R \to S$

◇ **Lemma 15.3.1.3** Let $R \to S$ be a ring map, M be an S-module. There is a canonical short exact sequence

$$0 \to \Omega_{S/R} \otimes_S M \to P^1_{S/R}(M) \to 0$$

functorial in M .

∘ **Proof** ... □

Tensor Product

♠ **Proposition 15.3.1.4** ...

∘ **Proof** ... □

15.4 The Naive Cotangent Complex

15.5 Smoothness

15.5.1 Formally Smooth Maps

Definition 15.5.1.1 ...

15.5.2 Smoothness and Differentials

15.5.3 Smooth Algebra over Fields

15.5.4 Smooth Ring Maps in the Noetherian Case

15.5.5 Formal Smoothness of Fields

15.6 Etale Ring Maps

Definition 15.6.0.1 Let $R \to S$ be a ring map. We say $R \to S$ is **etale** if it is

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15.7 Syntomic Morphisms

Chapter 16 Some of Rings

16.1 Local complete intersection Rings

Definition 16.1.0.1 Let k be a field, S be a finite type k-algebra:

- We say thta S is a **global complete intersection** over k if there exists a presentation

$$S = k[x_1, \dots, x_n]/(f_1, \dots, f_c)$$

  such that: $dim(S) = n - c$.

- We say that S is a **local complete intersection** over k if there exists a covering

$$Spec(S) = \cup D(g_i)$$

  such that each of the ring D_g is a global intersection over k.

◇ **Lemma 16.1.0.2** ...

16.1.1 Properties

16.1.2 Topology and Homology

16.1.3 Normal Bundle

16.1.4 Cohen-Macaulay Property

16.2 Gorenstein Rings

16.3 Regular Local Rings

16.4 Cohen-Macaulay Rings, Modules

16.5 Universal Catenary Rings

16.6 Henselian Local Rings

Chapter 17 Etale Map

Chapter 18 Japanese Rings, Nagata Rings

Chapter 19 Groebner Bases

Problems

- The ideal description problem: Does every ideal $I \subseteq k[X_1, \dots, X_n]$ has a finite generating set

$$\oplus k[X_1, \dots, X_n]f_i \to I \to 0 \text{ is exact} \quad \text{or} \quad I =< f_1, \dots, f_r >$$

- The ideal membership problem. Given $f \in K[X_1, \dots, X_n]$ and an ideal $I =< f_1, \dots, f_s >$. Determine if

$$f \in I$$

  or weaker version, we should determine if $V(f_1, \dots, f_s) \subseteq V(f)$.

- Solving the polynomial equation

$$\begin{cases} f_1(X_1, \dots, X_n) = 0 \\ \dots \\ f_s(X_1, \dots, X_n) = 0 \end{cases}$$

  Equivalently, asking the points in affine variety $V(f_1, \dots, f_s)$.

- The Implicitizatopm Problem: Let V be a subset of k^n, parametrically is

$$x_1 = g_1(t_1, \dots, t_m)$$
$$\dots$$
$$x_n = g_n(t_1, \dots, t_m)$$

  If: g_i is polynomial/rational function. Then, V is a variety/part of one variety. Find a family of polynomial equations defines the variety.

Consider some fundamental cases:

♣ **Example 19.0.0.1** $n = 1$: Then, $K[X]$ is PID. It has Eucilidean algorithm, to show

$$I = (g)$$

Those problems has a simple description.

♣ **Example 19.0.0.2** Degree $deg(f_i) = 1$, linear algebra problem.
Also the implicitization problem, $deg(g_i) = 1$.

19.1 Ordering of Monomials

Take the following notation

$$x^\alpha = x^{\alpha_1} \dots x^{\alpha_n}; \alpha \in \mathbb{Z}^n_{\geq 0}$$

we want to define an ordering, such that

$$x_i^{m+1} > x_i^m \qquad x_1 > \cdots > x_n$$

And the ordering is totally ordered. One and only one of

$$x^\alpha > x^\beta \quad x^\alpha = x^\beta \quad x^\alpha < x^\beta$$

should be true.
Operations: $x^\alpha > x^\beta \Rightarrow x^\alpha x^\gamma > x^\beta x^\gamma$.

Definition 19.1.0.1 A **monomial ordering** on $k[x_1, \ldots, x_n]$ is any relation $>$ on $\mathbb{Z}^n_{\geq 0}$, or on x^α, which satisfies

1. $>$ is total order;
2. $\alpha > \beta \Rightarrow \alpha + \gamma > \beta + \gamma$;
3. $>$ is well-ordered.

◇ **Lemma 19.1.0.2** $>$ is well-ordering iff every strictly decreasing sequence in $\mathbb{Z}^n_{\geq 0}$

$$\alpha(1) > \alpha(2) > \ldots$$

eventually terminates.

- **Proof**
- $\Rightarrow$
- $\Leftarrow$

□

Definition 19.1.0.3 (Lexicographic Order) Let $\alpha, \beta \in \mathbb{Z}^n_{\geq 0}$. We say:

$$\alpha >_{lex} \beta \Leftrightarrow \beta - \alpha \in \mathbb{Z}^n, \text{ the leftmost nonzero element is positive}$$

♠ **Proposition 19.1.0.4** The lex ordering on $\mathbb{Z}_{\geq 0}$ is a monomial ordering.

- **Proof** □

Definition 19.1.0.5 (Graded Lex Order)

Definition 19.1.0.6 (Graded Reverse Lex Order)

Definition 19.1.0.7 Let $f = \sum_\alpha a_\alpha x^\alpha$ be a nonzero polynomial in $k[x_1, \ldots, x_n]$. And let $>$ be a monomial order:

-
-
-

•

◇ **Lemma 19.1.0.8** Let $f, g \in k[X_1, \dots, X_n]$ be nonzero polynomials. Then:

- $Multideg(fg) = multideg(f) + multideg(g)$;
- If $f + g \neq 0$. Then, $multideg(f + g) \leq \max multideg(f), multideg(g)$

∘ **Proof** □

19.2 A Division Algorithm for Polynomial Rings $k[X_1, \dots, X_n]$

19.3 Hilbert's Nullstellensatz, and Groebner's Base

19.4 Properties of Groebner's Bases

19.5 Buchberger's Algorithm

Chapter 20 Elimintation Theory

Chapter 21   Reflexive Module

Chapter 22 Torsion

22.1 Torsion Module

Definition 22.1.0.1 Let R be a ring, M be an R-module:

- $I \subseteq R$ be an ideal. We say M is **I-power torsion module** if for any $m \in M$, there exists $n > 0$ such that
$$I^n m = 0$$
- Let $f \in R$. M is an **f-power torsion** module if for each $m \in M$, there exists an $n > 0$ such that
$$f^n m = 0$$
- Use the following notion:
$$M[I^n] := \{m \in M | I^n m = 0\} \qquad M[I^\infty] = \cup M[I^n]$$

Take the following notion $M[I^\infty] = \cup M[I^n]$.

◇ **Lemma 22.1.0.2** Let R be a ring. Then, let I be an ideal of R. Let M be an I-power torsion module. Then, M admits a resolution
$$\cdots \to K_2 \to K_1 \to K_0 \to M \to 0$$
with K_i is a direct sum of copies of R/I^n in n variable.

∘ **Proof** Take the smallest n_m such that $I^{n_m} \cdot m = 0$. Then, the surjection
$$\oplus_{m \in M} R/I^{n_m} \to M \to 0$$
is well-defined. Then take the digram $K_0 = Ker(\oplus_{m \in M} R/I^{n_m} \to M)$. □

♠ **Proposition 22.1.0.3** Let R be a ring. Let I be an ideal of R. Then, for any R-module M set
$$M[I^n] := \{m \in M | I^n m = 0\}$$
If I is finitely generated, then the followings are equivalent:

- $M[I] = 0$;
- $M[I^n] = 0; n \geq 1$;
- If $I = (f_1, \ldots, f_t)$, then the map $M \to \oplus M_{f_i}$ is injective.

∘ **Proof** (1)⇔ (2) Omitted.
(2)⇔ (3) We use the follwing lemma: □

♠ **Proposition 22.1.0.4** Let R be a ring. Let I be a finitely generated ideal of R:

- For any R-module M, we have
$$(M/M[I^\infty])[I] = 0$$

- An extension of I-power torsion modules is I-power torsion.

∘ **Proof**

-
- Consider the following extension:

$$...$$

□

22.1.1 Serre Subcategory

◇ **Lemma 22.1.1.1** ...

∘ **Proof** ... □

22.1.2 Cohomology

♠ **Proposition 22.1.2.1** ...

∘ **Proof** ... □
